BeyondContact: A Reinforcement Learning Framework for Airflow-Based Non-Contact Manipulation

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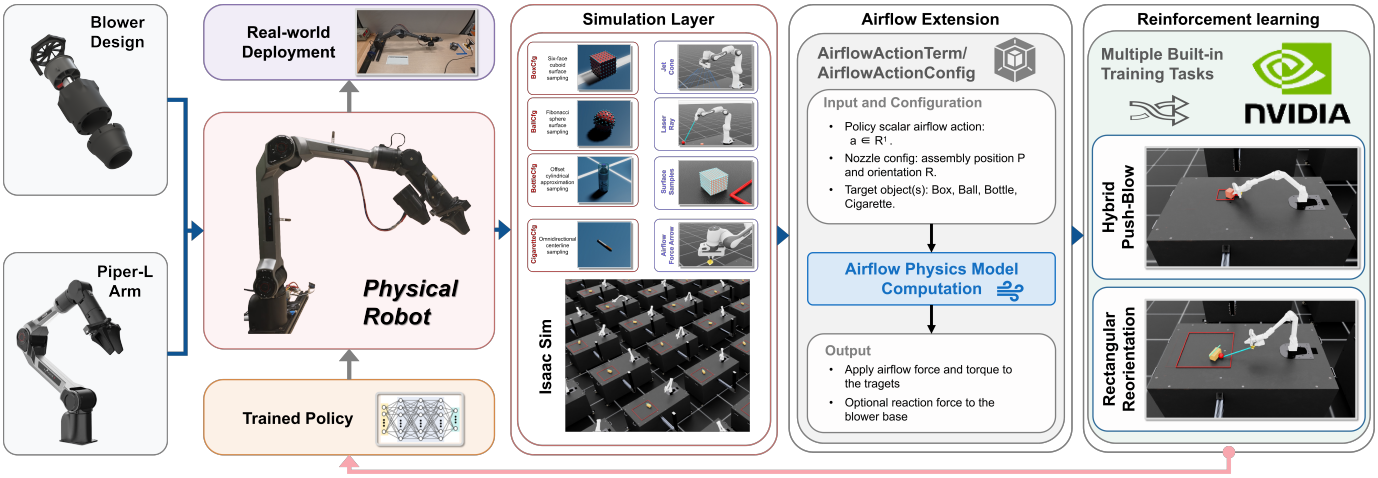

Fig. 1. Overview of the proposed reinforcement learning (RL) framework for airflow-based non-contact manipulation. A custom blower is integrated with a Piper-L robotic arm. The airflow simulation extension supports multiple object types and nozzle configurations, computing and applying airflow-induced forces and torques in simulation. The extension enables RL across multiple tasks, including hybrid push–blow (HPB) and rectangular reorientation (RR), with the trained policies subsequently deployed on the physical robot. The project webpage is available at: **https://anonymous.4open.science/r/BeyondContact-Anon/**

***Abstract*—Contact-based robotic manipulation is constrained by end-effector reachability and the mechanical properties of objects. Airflow can exert forces without physical contact, enabling manipulation of objects beyond the workspace or when direct contact is undesirable. However, reinforcement learning frameworks that incorporate airflow property into complex robotic systems remain underexplored. To address this gap, we design a lightweight, high-speed blower that coexists with a gripper and develop a robot- and task-agnostic airflow extension for Isaac Lab that supports massively parallel training. The extension uses experimentally identified airflow models to compute the forces and torques exerted on objects. Based on this extension, we train a hybrid push–blow (HPB) policy that combines contact-based pushing with airflow manipulation and a rectangular reorientation (RR) policy that controls a cuboid's heading using airflow alone. In zero-shot real-world deployment, HPB achieves an 80% success rate over 20 trials, while RR achieves a mean absolute heading error of 2.60° across four target headings.**

***Index Terms*—Reinforcement learning, grippers and other end-effectors, non-contact manipulation.**

I. Introduction

RECENT advances in robotic arm hardware and control methodologies have broadened the application of robotic manipulation in industrial [1] and domestic settings [2], [3]. However, prevailing manipulation paradigms remain predominantly based on physical contacts, such as grasping [4] and pushing [5], [6]. Although contact-based manipulation is effective in conventional tasks, including assembly, pick-and-place, and object transport, its applicability is constrained by the robot's kinematic workspace, collision-free reachability, and the mechanical properties of the object. In contrast, non-contact manipulation enables robots to act on objects without physical contact, offering the potential for safer operation and remote manipulation [7]. Among the available non-contact modalities [8]–[10], airflow is particularly attractive because it can be generated using compact hardware and can extend the action range beyond the reach of conventional end effectors while acting on multiple objects simultaneously.

Prior studies have explored airflow-assisted manipulation using diverse robotic platforms and control strategies. Lee *et al.* [11] and Natsu *et al.* [12] mounted fixed or steerable blowers on mobile platforms and used manual teleoperation or predefined coverage trajectories for large-area leaf collection and cleaning. Wu *et al.* [13] investigated reinforcement learning (RL)-based control of a mobile blower, approximating airflow–object interactions using virtual particles in simulation. However, their representation does not explicitly model the spatial decay and spread of the airflow velocity field, and their evaluation was limited to planar blowing tasks without considering 3D nozzle pose control.

In [14] and [15], the researchers developed a compressed-air system with a two-degree-of-freedom (DoF) nozzle mounted at a fixed height to analyze the jet-induced airflow field over a planar surface. The former employed a heuristic proportional feedback controller to manipulate objects of diverse shapes at meter-scale distances, whereas the latter combined identified object dynamics with a cross-entropy method (CEM)-based controller to perform path following, aggregation, and sorting of single and multiple objects. Both studies relied on prede-

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fined feedback laws or model-based optimization rather than policies learned through RL.

DextAIRity [16] is closely related to our work, equipping one arm of a three-arm system with a commercial blower for deformable object manipulation. It uses a particle-based airflow model to train cloth-unfolding policies in simulation. However, this simplified model is insufficient for bag-opening policy training, which relied on 4,400 real-world interactions collected over 10 hours. Moreover, the blower arm lacks a gripper, requiring separate arms for grasping.

Additionally, alternative airflow actuation mechanisms have been explored: Hyungpil *et al.* [17] proposed manipulating planar objects using two airflow sinks; Artur *et al.* [18] achieved active 3D space manipulation of a lightweight airborne origami object using patterned airflow generated by a 3×3 array of computer fans; and Bao *et al.* [19] developed a Bernoulli-based non-contact robotic manipulator, in which a nozzle array directs compressed air toward the object surface to create a low-pressure region and hold the object.

Despite the demonstrated potential of airflow-based manipulation, a computationally efficient airflow–object interaction module for massively parallel RL remains underexplored. Such a module, supported by systematic parameter identification and experimental validation, would facilitate adaptation to diverse robotic platforms and custom blower configurations, policy training in simulation, and sim-to-real deployment.

To address this gap, we present an airflow-interaction extension for Isaac Lab. The extension incorporates a configurable airflow-field model characterized by a centerline-decay function and a spatially varying radial profile, mapping local airflow velocities to forces and torques applied to simulated rigid bodies of various shapes. It can also be adapted to different robotic platforms and nozzle configurations while maintaining computational efficiency for parallel RL training. Building upon this extension, we present two RL training paradigms for the blower-equipped Piper-L robotic arm: hybrid push–blow (HPB), which coordinates contact pushing and non-contact airflow actuation to transport objects to target regions beyond the arm's direct reach, and rectangular reorientation (RR), which uses airflow alone to align objects with target orientations. Finally, the learned policy is deployed and evaluated on a real robot.

To summarize, the main contributions of this work are:

- Develop a configurable and computationally efficient airflow-interaction module used in Isaac Lab, supporting massively parallel robot-learning environments.
- Design a lightweight, high-speed blower that coexists with the gripper, delivering a maximum air velocity of $78\,\mathrm{m/s}$ and a static thrust of up to $5.9\,\mathrm{N}$ ($600\,\mathrm{gf}$).
- Experimentally identify centerline-decay and radial-spreading models for custom blowers and nozzles.
- Propose RL training frameworks that enable hybrid contact–airflow manipulation in HPB and purely airflow-driven orientation control in RR, and validate the learned policies through zero-shot sim-to-real deployment.

The airflow extension, RL frameworks, and blower design are available on the project webpage for reproducible research: https://anonymous.4open.science/r/BeyondContact-Anon.

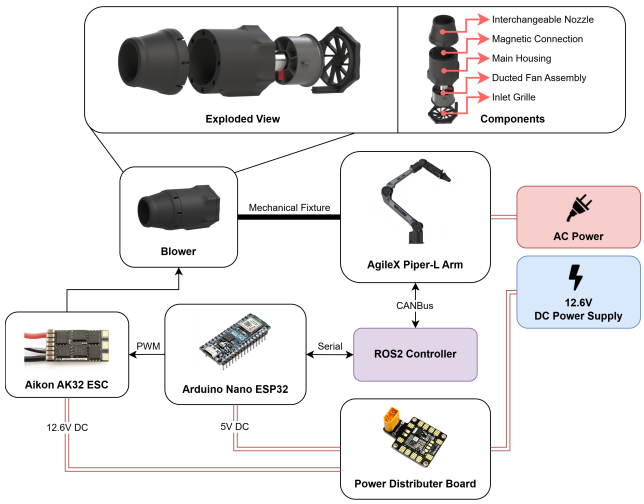

Fig. 2. Exploded view of the custom blower and system architecture showing control interfaces and power distribution.

II. Mechatronic Design

We designed a compact blower module that can be mounted on the end effector of the Piper-L robotic arm, directly below the gripper. The module houses a $40\,\mathrm{mm}$ *XFly Galaxy X4* electric ducted fan (EDF) in a 3D-printed enclosure. The EDF is driven by an *Aikon AK32* electronic speed controller (ESC) and can be powered by an external supply within the nominal voltage range of 3S–4S batteries. An *Arduino Nano ESP32* generates PWM signals to regulate the blower output. All connections are summarized in the circuit diagram in Fig. 2.

N52 neodymium magnets provide a detachable connection at the blower outlet to accommodate different nozzle configurations. In this work, we provide a nozzle design with an outlet diameter of $31.7\,\mathrm{mm}$ and an outlet area corresponding to 80% of the EDF fan-swept area.

III. Airflow Dynamics Modeling

This section presents the airflow model. We represent the blower-generated airflow as a time-averaged, quasi-steady free turbulent round jet in a quiescent environment, capturing the mean velocity field across the near-field, transitional, and self-similar far-field regions while maintaining computational efficiency [20]. The flow is assumed incompressible and axisymmetric about the jet centerline [21], [22]. Rotor-induced swirl, instantaneous turbulent fluctuations, and interactions with nearby environmental surfaces are not explicitly resolved.

A. Nozzle-Centered Representation

Let $\mathbf{p}_n$ be the nozzle center and $\hat{\mathbf{e}}_n$ be the unit vector along the nozzle axis. The axial and radial coordinates of a spatial point $\mathbf{p}$ in the nozzle-centered coordinate system are defined as:

$$x = (\mathbf{p} - \mathbf{p}_n)^\top \hat{\mathbf{e}}_n, \tag{1}$$

$$r = \left\| (\mathbf{p} - \mathbf{p}_n) - x\hat{\mathbf{e}}_n \right\|_2 . \tag{2}$$

The above model applies to the downstream ($x > 0$). Assuming a non-swirling turbulent free round jet, the mean

transverse velocity components U_y and U_z in the nozzle coordinate system are negligible with respect to U_x [23], and the local airflow velocity vector is given by Eq. (3):

$$\mathbf{u}_{\text{air}}(\mathbf{p}) = U_x(x, r)\hat{\mathbf{e}}_n = U(x, r)\hat{\mathbf{e}}_n, \tag{3}$$

where $U(x, r)$ denotes the local airflow speed.

B. Centerline Jet Decay

Existing turbulent round jet models define an initial potential core followed by a developing region and a self-similar far field. The potential core is established by a region that ranges approximately between $4D$ and $6D$, where D is the nozzle outlet diameter, and the centerline velocity remains identical or close to the outlet velocity U_0. Further downstream beyond the near-field potential core, the airflow transitions into a self-similar jet, which is approximately modeled as an inverse relationship between axial velocity and centerline distance from the jet outlet.

Following this modeling approach, we formulate the centerline velocity as given in Eq. (4):

$$U_c(x) = U_0 \cdot f_{\text{dec}}(x), \tag{4}$$

where $f_{\text{dec}}(x)$ is the dimensionless decay function associated with model m. We consider two candidate models in this work: an exponential model and a shifted inverse model.

The exponential model is defined as:

$$f_{\text{exp}}(x) = \begin{cases} 1, & 0 < x \leq x_c, \\ \exp[-k(x - x_c)], & x > x_c, \end{cases} \tag{5}$$

where x_c is the aforementioned potential core length and $k > 0$ is the decay coefficient. This model provides a bounded representation and forms a compact empirical approximation within a finite operating range. Moreover, the model does not seek to replicate the asymptotic properties of a fully turbulent jet because it tends to zero faster than an inverse power-law function.

The shifted inverse model is inspired by the Albertson three-zone model [20] and is defined as:

$$f_{\text{inv}}(x) = \min(B \cdot \frac{D}{x - x_0}, 1) \tag{6}$$

where x_0 is the virtual origin and B is the dimensionless decay coefficient. The $\min$ function in Eq. (6) ensures that the model remains bounded at the nozzle.

C. Radial Jet Spreading

Following the Albertson three-zone model, we model the radial distribution of the mean axial velocity using a Gaussian profile:

$$U(x, r) = U_c(x) \cdot \exp\left[-\frac{r^2}{2\sigma_r^2(x)}\right], \tag{7}$$

where $\sigma_r(x)$ is a spatial jet width across the radial dimension, and is modeled approximately as a linear spreading of the turbulent round jet:

$$\sigma_r(x) = a_\sigma + Sx, \tag{8}$$

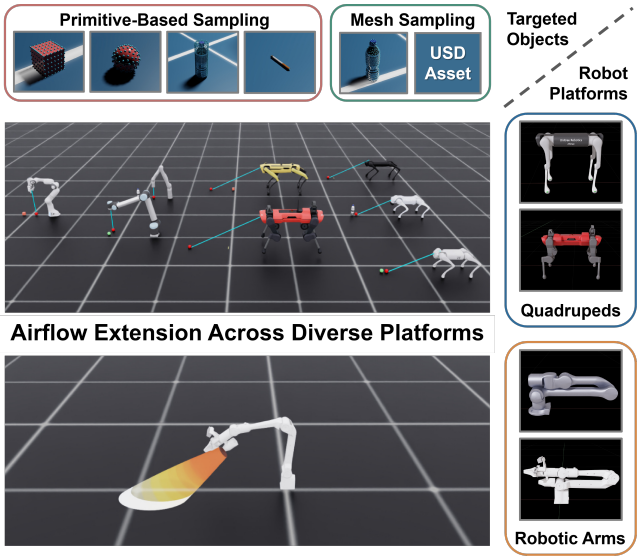

Fig. 3. Demonstration of the airflow extension across diverse target objects and robotic platforms. Surface primitives are sampled from basic geometric shapes or mesh geometry in USD assets, while blowers can be mounted on various platforms, including quadrupeds and robotic arms.

where a_σ is the extrapolated width intercept and S is the dimensionless spreading rate. The intercept accounts for the finite initial jet width and avoids forcing the spreading line through the nozzle origin.

Consequently, Eqs. (4)–(8) describe axial decay and radial spreading separately. All the model parameters, including the reference exit velocity U_0, centerline-decay parameters, and radial-spreading parameters, are identified from experimental measurements of the blower with and without a nozzle.

IV. Airflow Extension

This section presents the implementation of the airflow extension and its integration into Isaac Lab for large-scale robot learning. The extension maps the airflow dynamics model established in Sec. III to forces and torques acting on target objects through batched PyTorch tensor operations. These forces and torques are applied through the Isaac Sim/PhysX physics backend to simulate object motion and scene interactions. The extension is robot- and task-agnostic, accommodating diverse airflow configurations, target objects, and learning environments. Fig. 3 illustrates its application to various robotic platforms and target objects.

Specifically, the extension supports surface sampling for cuboids, spheres, cylinders, and arbitrary mesh-based objects, and provides four ready-to-use object models: a cube, a sphere, a plastic bottle, and a cigarette. During initialization, the surface of each user-specified target object is decomposed once into a set of primitives:

$$\mathcal{P} = \{\mathbf{p}_i, \mathbf{n}_i, A_i\}, \tag{9}$$

where $\mathbf{p}_i$ and $\mathbf{n}_i$ denote the position and outward surface normal of the i-th primitive expressed in the object frame, respectively, and $\mathbf{A}_i$ is its associated surface area.

The local airflow velocity $\mathbf{u}_i$ at each primitive is computed using the airflow model. Following [24], the corresponding aerodynamic force is then calculated as:

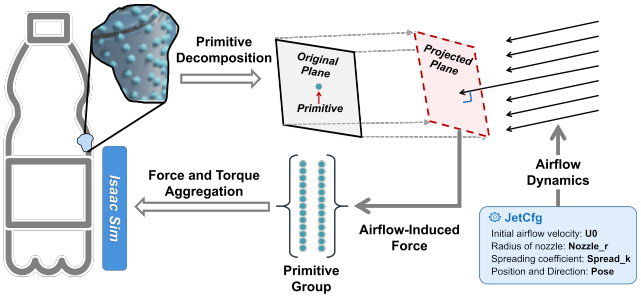

Fig. 4. Force computation pipeline of the airflow extension. The object's surface is decomposed into primitives, and local airflow-induced forces are computed based on the airflow configuration and each primitive's projected area. These forces are then aggregated into the net force and torque on the object and applied in Isaac Sim.

$$\mathbf{F}_i = \frac{1}{2}\rho C_D \left\|\mathbf{u}_i\right\|_2^2 A_i \left[-\hat{\mathbf{e}}_n^T \mathbf{n}_i\right]_+ \hat{\mathbf{e}}_n, \tag{10}$$

where ρ is the air density, C_D is the effective drag coefficient, and $[\cdot]_+$ denotes the positive operator. In this equation, only primitives facing the incoming airflow contribute to the resulting force.

By integrating over all surface primitives, the force acting on the target object and the torque about its center of mass (CoM) are computed as:

$$\mathbf{F} = \sum_i \mathbf{F}_i, \qquad \boldsymbol{\tau} = \sum_i (\mathbf{p}_i - \mathbf{p}_o) \times \mathbf{F}_i, \tag{11}$$

where $\mathbf{p}_o$ denotes the CoM of the target object.

The force and torque are applied to the target rigid body through the external-force interface of Isaac Lab. Compared with a simplified airflow force evaluated only at the object center, the proposed surface-integration method captures the eccentric torque induced by object orientation and asymmetric airflow coverage. The extension also provides an interface for applying a reaction force, scaled with the airflow action, to the rigid body carrying the blower. This approximates the disturbance exerted by the high-speed airflow on the robot link and further reduces the sim-to-real gap. The computation pipeline is shown in Fig. 4.

V. Training Framework

A. Task Formulation

To evaluate the proposed framework using a Piper-L robot arm[1], we design two complementary tasks to evaluate the proposed framework: a HPB task and a RR task.

In the HPB task, the policy moves an object from a random initial position to a random 2D target region to evaluate manipulation beyond the arm's direct reach. In the RR task, the policy rotates a cuboid from a random initial orientation to the target orientation while keeping it within the working area, to evaluate airflow-based orientation control.

Both tasks are formulated as Markov decision processes (MDPs):

$$\mathcal{M}^{(k)} = \left(\mathcal{S}^{(k)}, \mathcal{A}, p, r^{(k)}, \gamma\right), k \in \{\mathrm{HPB}, \mathrm{RR}\}, \tag{12}$$

[1]Compared with 7-DoF arms, the Piper-L robotic arm constrains the achievable combinations of nozzle position and orientation, limiting the range of manipulation achievable through airflow alone.

TABLE I
Key PPO hyperparameters shared by the two tasks.

Parameter	Value	Parameter	Value
Learning rate	3×10^{-4}	Clipping	0.2
Discount factor, γ	0.99	GAE parameter, λ	0.95
Epochs	6	Mini-batches	8
Value-loss coefficient	1.0	Entropy coefficient	0.002
Gradient norm	1.0	Initial STD	1.0

where $\mathcal{S}^{(k)}$ and $\mathcal{A}$ denote the task-specific state space and the shared action space, respectively; p denotes the environment transition dynamics; $r^{(k)}$ is the task-specific reward function; and γ is the discount factor. At each control step, the policy outputs control commands for the robot arm and blower according to the current observation and task goal.

For each task, the policy is optimized by maximizing the expected discounted return:

$$\max_\theta \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T} \gamma^t r_t^{(k)}\right]. \tag{13}$$

Proximal policy optimization (PPO) is adopted to train the policies. The two tasks share the same training framework and optimization method, with the PPO hyperparameters summarized in Table I. The actor–critic structures, reward design, and domain randomization are described in the following subsections.

B. Asymmetric Actor–Critic Architecture

Both tasks are trained using an asymmetric actor–critic architecture. Both the actor and critic are implemented as three-hidden-layer MLPs with ELU activations and hidden dimensions of $[512, 256, 128]$.

The actor takes observations available during deployment as input:

$$\mathbf{o}_t^{\mathrm{actor}} = \left[\mathbf{q}_t^{\mathrm{hist}}, \dot{\mathbf{q}}_t^{\mathrm{hist}}, \mathbf{x}_{b,t}, \mathbf{g}_t^{(k)}, \mathbf{a}_{t-1}\right], \tag{14}$$

where $\mathbf{q}_t^{\mathrm{hist}}$ and $\dot{\mathbf{q}}_t^{\mathrm{hist}}$ denote the 10-step histories of joint positions and velocities; $\mathbf{x}_{b,t}$ is the object pose; $\mathbf{g}_t^{(k)}$ is the task goal, i.e., a 2D target position for HPB and an additional 1D orientation for RR; and $\mathbf{a}_{t-1}$ is the previous actions.

The critic receives additional privileged information during training:

$$\mathbf{o}_t^{\mathrm{critic}} = \left[\mathbf{o}_t^{\mathrm{actor}}, \dot{\mathbf{x}}_{b,t}, \boldsymbol{\omega}_{b,t}, \Delta\mathbf{p}_{g,t}^{(k)}, R, \boldsymbol{\eta}_{n,t}\right], \tag{15}$$

where $\dot{\mathbf{x}}_{b,t}$ and $\boldsymbol{\omega}_{b,t}$ are the object's linear and angular velocities, $\Delta\mathbf{p}_{g,t}^{(k)}$ is the object-to-goal displacement, R denotes the target tolerance, and $\boldsymbol{\eta}_{n,t}$ contains the nozzle-axis direction and its horizontal and downward components.

C. Reward Design

The task-specific total reward is defined as:

$$r_{\mathrm{total}}^{(k)} = r_{\mathrm{task}}^{(k)} \cdot \Delta t + r_{\mathrm{penalty}} \cdot \Delta t + r_{\mathrm{regularization}} \cdot \Delta t, \tag{16}$$

where $r_{\mathrm{task}}^{(k)}$ denotes the task-level reward, $r_{\mathrm{regularization}}$ denotes the regularization reward, r_{penalty} denotes the penalty terms, and Δt is the control interval. The detailed rewards are summarized in Table II, and the design of each component is described in the following subsections.

TABLE II
REWARD FUNCTION TERMS AND WEIGHTS.

Reward term	Formula	Weight
HPB Task-level Rewards		
Dense goal reward	$1-\tanh(d_t/\sigma_d)$	2.0
Goal progress	$(d_{t-1}-d_t)/\sigma_d$	3.0
Goal inside	$\mathbb{I}_{\text{in}}$	8.0
Goal stable	$\mathbb{I}_{\text{in}}\exp(-\lVert\mathbf{v}_{b,xy}\rVert_2^2/\sigma_v^2)$	2.0
Nozzle–object range	$\exp(-(d-d^*)^2/\sigma_r^2)$	0.05
Airflow toward goal	$\max(\mathbf{v}_{b,xy}^T\hat{\mathbf{e}}_g,\,0)u_t g_{\text{ready}}$	6.0
Directed airflow force	$\mathbb{I}_{\text{act}}\,\text{clip}(F_g/F_s,0,1)$	1.0
RR Task-level Rewards		
Dense goal reward	$1-\tanh(d_t/\sigma_d)$	2.0
Goal progress	$(d_{t-1}-d_t)/\sigma_p$	3.0
Goal inside	$\mathbb{I}_{\text{in}}$	2.0
Orientation alignment	$g_{\text{area}}(1-2\delta_t/\pi)\mathbb{I}_{\text{valid}}$	1.0
Orientation progress	$g_{\text{area}}(\delta_{t-1}-\delta_t)\mathbb{I}_{\text{valid}}/\sigma_\theta$	2.0
Pose achieved	$\mathbb{I}_{\text{pose}}$	4.0
Pose stable	$\mathbb{I}_{\text{pose}}\exp\big(-\lVert\mathbf{v}_{b,xy}\rVert_2^2/\sigma_v^2-\lVert\boldsymbol{\omega}_{b,z}\rVert_2^2/\sigma_\omega^2\big)$	2.0
Nozzle–object range	$\exp(-(d-d^*)^2/\sigma_r^2)$	0.05
Airflow toward goal	$\max(\mathbf{v}_{b,xy}^T\hat{\mathbf{e}}_g,\,0)u_t g_{\text{ready}}$	6.0
Directed airflow force	$\mathbb{I}_{\text{act}}\,\text{clip}(F_g/F_s,0,1)$	1.0
Shared Regularization		
Airflow energy	$s_{\text{air}}u_t^2$	-0.08
Airflow rate	$s_{\text{air}}(u_t-u_{t-1})^2$	-0.08
Arm action deviation	$\lVert\mathbf{q}_t^{\text{cmd}}-\mathbf{q}_0\rVert_2^2$	-0.005
Arm action rate	$\lVert\mathbf{a}_t^{\text{arm}}-\mathbf{a}_{t-1}^{\text{arm}}\rVert_2^2$	-0.08
Joint velocity	$\lVert\dot{\mathbf{q}}_t\rVert_2^2$	-0.03
End-effector acceleration	$s_{\text{ee}}\lVert\mathbf{a}_t^{\text{ee}}\rVert_2^2$	-0.002
Shared Penalty		
Airflow before ready	$s_{\text{air}}u_t^2(1-Q_{\text{ready}})$	-0.40
Nozzle too low	$s_{\text{air}}u_t d_{\text{low}}^2$	-0.10
Downward nozzle pitch	$s_{\text{air}}u_t\theta_{\text{down}}/\theta_0$	-0.50
Robot contacts	N_{contact}	-2.0
Object out of workspace	$\mathbb{I}[\mathbf{p}_b\notin\mathcal{W}]$	-200.0
Object tumbling	$\mathbb{I}_{\text{tumble}}$	-50.0

Notations: d: distances; σ, F_s, θ_0: normalization scales; $\mathbb{I}$: condition indicators; $\mathbf{p}, \mathbf{v}, \boldsymbol{\omega}$: position, linear/angular velocities; $\hat{\mathbf{e}}_g$: goal direction; u: airflow command; g: reward gates; F_g: goal-directed airflow force; δ: heading error; s: penalty scales; $\mathbf{q}, \dot{\mathbf{q}}$: joint positions/velocities; $\mathbf{a}$: accelerations. Q_{ready}: nozzle readiness; θ_{down}: downward nozzle pitch; N_{contact}: undesired contact; $\mathcal{W}$: workspace.

1) Hybrid Push–Blow Task-level Reward: This task reward guides the object toward the target region while promoting effective airflow manipulation. Dense and progress rewards encourage goal-directed motion, whereas inside and stability rewards encourage the object to enter the target region and remain stationary. The nozzle–object range reward promotes an effective airflow distance. The airflow toward goal reward encourages goal-directed object motion when the blower is activated and is modulated by g_{ready}, which evaluates nozzle placement and axis direction relative to the object motion.

2) Rectangular–Reorientation Task-level Reward: Building on the HPB task-level rewards, this reward set further introduces dense and progress rewards for orientation alignment. Pose achievement and stability rewards encourage the object to remain stationary within the target region while maintaining the desired orientation. The nozzle–object range and airflow toward goal rewards are also retained.

3) Shared Regularization and Penalties: The regularization terms promote smooth and efficient policy execution. The airflow energy and rate terms penalize excessive airflow intensity and rapid changes in airflow, respectively, while the action-rate and joint-velocity terms constrain abrupt changes in the arm control policy and aggressive joint motions. Once the object enters the target region, the arm action deviation term further encourages the robot arm to return to its default posture.

TABLE III
DOMAIN RANDOMIZATION PARAMETERS.

Parameter	Range
Object mass	$[0.8,\ 1.2]\times m_{\text{design}}$
Static friction	$[0.8,\ 1.2]$
Dynamic friction	$[0.7,\ 1.1]$
Robot stiffness	$[0.8,\ 1.2]$
Robot damping	$[0.8,\ 1.2]$
Outlet velocity, U_0	$[24.5,\ 30.0]$ m/s
Spreading coefficient, k	$[0.0688,\ 0.0840]$

TABLE IV
CURRICULUM SETTINGS FOR HPB AND RR. STEP COUNTS ARE EXPRESSED IN THOUSANDS (K).

Curriculum	Initial	Final	Update
HPB Task Curriculum			
Target half-width (m)	0.12	0.10	Success rate
Target x-range (m)	$[0.55, 0.75]$	$[0.40, 1.40]$	Success rate
Target y-range (m)	$[-0.08, 0.08]$	$[-0.15, 0.15]$	Success rate
Air penalty, s_{air}	0	1	5–15k steps
RR Task Curriculum			
Heading tolerance	30°	10°	Success rate
Success hold (s)	0.50	1.00	Success rate
Heading half-range	20°	45°	Success rate
Air penalty, s_{air}	0	1	30–80k steps

The penalty terms discourage airflow activation when the nozzle is not correctly positioned, too low, or at a large downward angle. Penalties also apply to undesired robot contacts, object motion outside the workspace, and object tumbling.

4) Domain Randomization, Termination, and Curriculum: To reduce the sim-to-real gap arising from modeling and control uncertainties, object properties, robot actuation gains, and the identified airflow parameters are randomized during training, as summarized in Table III.

An episode terminates upon timeout, object departure from the workspace[2], an excessively low end-effector height, or collision between the blower and the object. For the RR task, we also add gripper-object contact as a termination condition.

Table IV summarizes the curriculum used in both HPB and RR policy training. Task difficulty and penalty scales are gradually increased to facilitate learning.

VI. EXPERIMENTAL RESULTS

This section presents the experimental setup and results. Airflow identification experiments determine the model parameters and evaluate model accuracy. Real-world deployment experiments were conducted to assess the performance of the trained policies.

A. Airflow Identification

1) Numerical and Experimental Setup: To establish a comparison baseline, we simulate an idealized turbulent round jet in OpenFOAM v14 [25], solving the incompressible RANS equations with the standard k-ϵ model and SIMPLE algorithm. The axisymmetric 5° wedge extends $50D$ axially and $15D$ radially, with a 4:1 graded $200\times6\times1$ mesh of 12,000 cells. The inlet uses a top-hat profile with $U_{0,\text{EDF}} = 27.28\text{m/s}$ and $U_{0,\text{nozzle}} = 30.5\text{m/s}$, 5% turbulence intensity, and

[2]The workspace is defined as $x\in[0.05, 1.65]$ m, $y\in[-0.40, 0.40]$ m.

TABLE V
CENTERLINE MODEL-IDENTIFICATION RESULTS WITH MODEL PARAMETERS AND THEIR RESPECTIVE STANDARD ERROR (SE)

Dataset	Model	RMSE (m/s)	R^2	MARE (%)	$k \pm$ SE (m^{-1})	$B \pm$ SE (−)	$x_0 \pm$ SE (mm)
CFD ($D = 40$ mm)	Exponential	1.736	0.927	30.2	2.293 ± 0.044	—	—
	Inverse	**0.350**	**0.997**	**3.4**	—	3.324 ± 0.037	-36.1 ± 2.2
CFD ($D = 31.7$ mm)	Exponential	1.939	0.927	30.2	2.893 ± 0.055	—	—
	Inverse	**0.392**	**0.997**	**3.4**	—	5.324 ± 0.037	-28.6 ± 1.7
EDF without Nozzle	**Exponential**	**0.606**	**0.994**	**6.6**	2.648 ± 0.065	—	—
	Inverse	1.066	0.981	10.9	—	4.573± 0.279	15.9 ± 14.1
0.8-FSA Nozzle	Exponential	1.002	0.988	8.7	3.170 ± 0.077	—	—
	Inverse	**0.895**	**0.991**	**7.2**	—	4.565 ± 0.198	29.9 ± 9.7

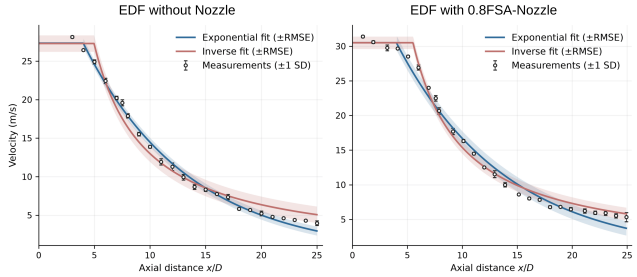

Fig. 5. Centerline velocity comparison for the no-nozzle and nozzle configurations. Axial distance is normalized by D.

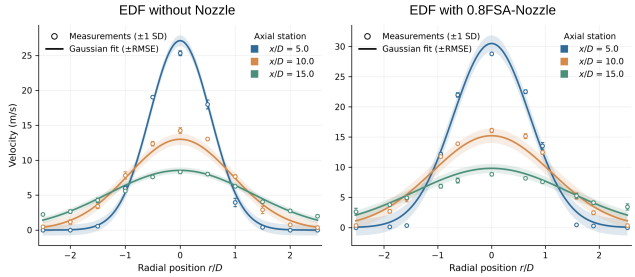

Fig. 6. Measured and fitted radial velocity profiles for no-nozzle and nozzle configurations. Radius is normalized by D, and color identifies the axial station x/D.

$1.516\times10^{-5}\mathrm{m}^2/\mathrm{s}$ kinematic viscosity. After 500 iterations, all initial residuals are below 10^{-4}. The model excludes motor-hub blockage and blade-induced swirl.

Airflow measurements were recorded at centerline and radial stations using the *HoldPeak HP-5000Y* anemometer ($\pm5\%$ accuracy). The EDF operated at 12.6V and 50% PWM, drawing approximately 1.89A. Three readings were taken per station after 5s of stabilization. Centerline measurements were obtained by translating the aligned blower along its axis, while radial measurements were taken in both directions at $x/D = 5, 10,$ and 15. The reference diameters were $D_{\mathrm{EDF}} = 40\mathrm{mm}$ and $D_{\mathrm{nozzle}} = 31.7\mathrm{mm}$.

2) Centerline Model Identification: Both centerline models were evaluated against CFD and experimental data at identical stations. The exponential model used $x_c = 4D$, while the shifted-inverse model was capped at U_0 near the outlet. Table V summarizes the errors for the CFD, EDF without nozzle, and with nozzle cases, while Fig. 5 compares the normalized measurements and fitted models.

Against CFD, the shifted-inverse model reduced the RMSE by 79.81% relative to the exponential model for both blower configurations, as expected for a developed turbulent round jet. For the bare EDF, however, the exponential model achieved a lower RMSE of $0.606\mathrm{m/s}$ versus $1.066\mathrm{m/s}$.

The 0.8-FSA nozzle produced a more axisymmetric round-jet-like flow, for which the shifted-inverse model achieved a lower RMSE of $0.895\mathrm{m/s}$ versus $1.002\mathrm{m/s}$. The maximum experimental RMSE and MAE were $1.066\mathrm{m/s}$ and $2.217\mathrm{m/s}$, respectively, partly reflecting the anemometer accuracy. Ultimately, the results favor the shifted-inverse model since it performs better overall, even though the exponential model remains a competitive approximation when it comes to capturing the centerline velocity decay of a free turbulent round jet.

3) Radial Profile Identification: Radial measurements at three centerline stations were fitted with Gaussian profiles. To decouple them from the centerline models, both the amplitude and σ_r were fitted directly to the experimental data. Opposite radial measurements differed by only 3%–7%, indicating approximate axisymmetry within the anemometer accuracy. The fits achieved $R^2 \geq 0.960$ and $R^2 \geq 0.970$ for the EDF with and without nozzle configurations, respectively. As shown in Fig. 6, σ_r increased approximately linearly with distance. The CFD, bare-EDF, and nozzle spreading rates were 0.0963, 0.0766, and 0.0828, corresponding to width angles of $5.50°$, $4.38°$, and $4.73°$, respectively.

B. Policy Deployment

1) Policy Training: We trained the policy using a model of the blower-equipped Piper-L arm identified with the open-source method in [26]. Training was conducted on a computer equipped with an NVIDIA RTX 5090 GPU and an Intel U7 265K CPU. The policies were trained using 4096 parallel environments for 5000 iterations, with a total training time of 5.7 h (HPB) and 6.6 h (RR). The trained policies achieved success rates of 88% on the HPB task and 92% on the RR task in simulation.

2) Experimental Setup: For real-world experiments, the Piper-L arm was mounted at the short edge of a $0.8 \times 1.8\,\mathrm{m}$ table, with its base 3 cm above the tabletop. A calibrated D435f camera was mounted on the table and used to detect the objects, while object-mounted AprilTags [27] enabled real-time orientation estimation. The gripper was foam-padded to cushion contact impacts.

3) Real-world Results:

a) HPB Task: In the HPB task, when the target region is set within the gripper's reach, the policy primarily uses contact-based manipulation, consistent with the airflow-use penalty. When the target region is beyond the gripper's reach, the policy first repositions the cube by pushing and then uses airflow to blow it into the target region (Fig. 7).

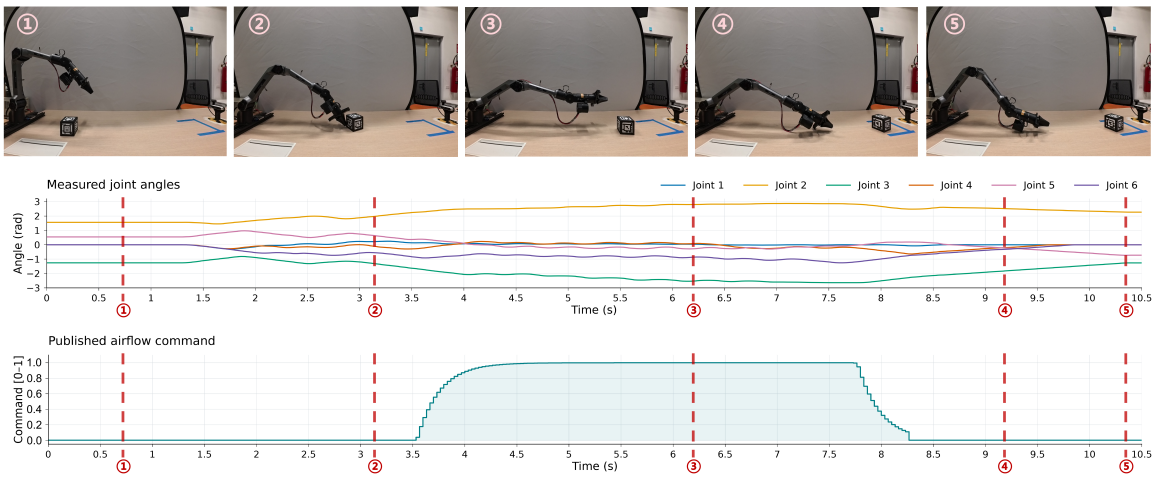

Fig. 7. HPB task policy deployment. The arm first pushes the cube into a position suitable for blowing (①–②). It then aims the blower at the cube, activates the airflow, and adjusts the blowing direction to guide the cube toward the target region (③). Once the cube enters the target region, the airflow command decreases to zero, leaving the cube in the target region (④–⑤). The plots below show the measured angles of the six joints and the airflow command; the red dashed lines mark the times corresponding to the images above.

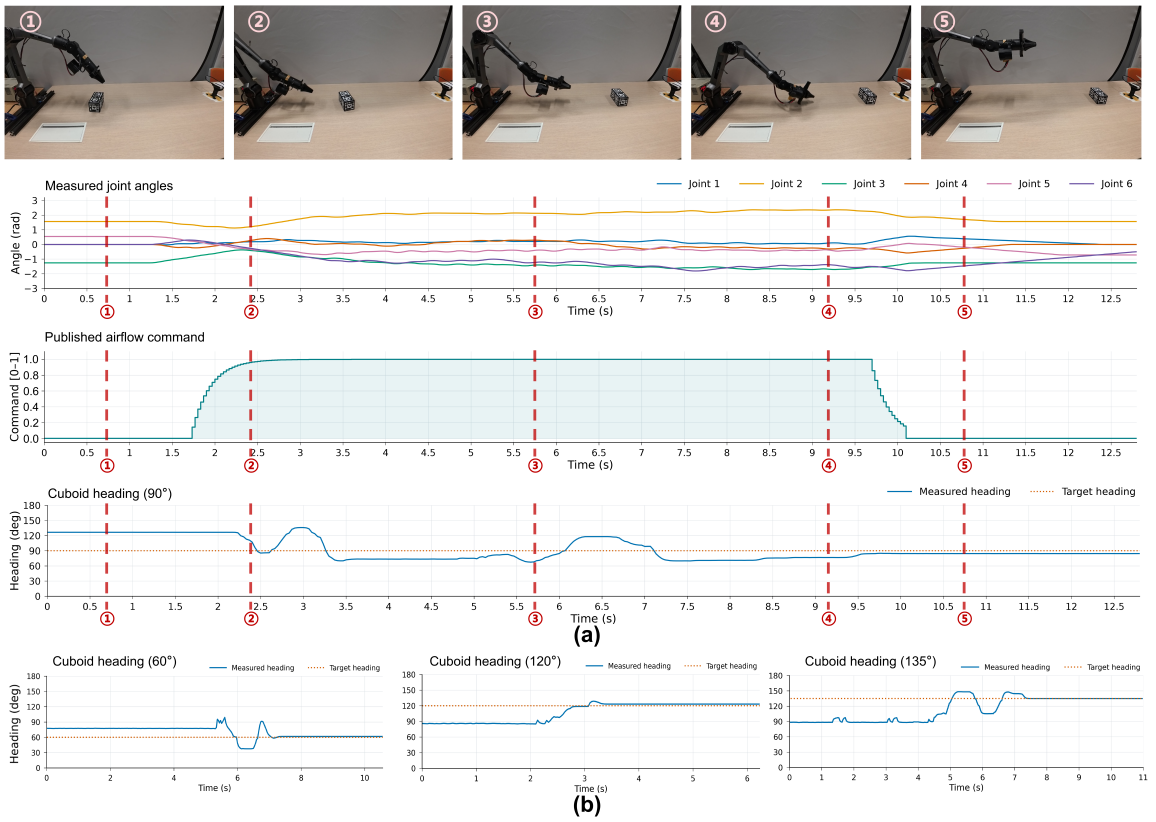

Fig. 8. RR task policy deployment. (a) For a target heading of 90°, the arm continuously adjusts the blowing direction to rotate the cuboid using contact-free airflow (①–④). Once the cuboid reaches the target heading, the airflow command gradually decreases to zero (⑤). The plots below show the measured angles of the six joints, the airflow command, and the cuboid heading; the red dashed lines mark the times corresponding to the images above. (b) Measured and target headings for target angles of 60°, 120°, and 135°.

Furthermore, to evaluate the policy's success rate over repeated trials, we defined a fixed square target region with a half-width of 10 cm, centered 1.3 m in front of the robot base, and conducted 20 trials. The cube successfully entered the target region in 16 trials, achieving a success rate of 80%.

Across the repeated trials, failures fell into two main categories: overly aggressive motions during pushing that displaced the cube beyond the range where airflow or pushing could correct it; and arm oscillations during blowing that caused unintended contact with the object, pushing it away from the correct direction. Both failures were contact-related. The arm's limited positional and orientational reach in the current setup made recovery difficult once the object deviated, resulting in task failure. These issues can be mitigated by using a 7-DoF arm or incorporating a mobile base.

b) RR Task: We conducted experiments with target of 60°, 90°, 120°, and 135° in the RR task. The policy manipulated the cuboid using airflow only, with the arm adjusting

the blowing direction to rotate the cuboid toward the target heading. Fig. 8(a) shows the deployment detail for the 90° trial, while Fig. 8(b) shows the heading trajectories for the other trials. The final headings were 61.58°, 84.47°, 123.18°, and 134.88°, all within the $\pm 10^\circ$ tolerance, and with a mean absolute error of 2.60°.

VII. Conclusion

This work designed a lightweight, high-speed blower that can be mounted on a robotic arm alongside its gripper, developed an Isaac Lab airflow-interaction extension for massively parallel RL training, and experimentally identified models of airflow decay and spreading. Building on these components, we proposed training frameworks for the HPB and RR tasks, enabling hybrid contact-based pushing and non-contact blowing in HPB and airflow-only object reorientation in RR. In zero-shot real-world deployment, HPB achieved an 80% success rate over 20 trials, while RR achieved a mean absolute heading error of 2.60° across four target headings.

Policy training and real-world deployment in the current system are constrained by the arm's kinematic reachability, limiting the positions and directions from which the blower can apply airflow to the target. Future work will explore adding a mobile base to the arm–blower system to expand its operating range, improve recovery after disturbances, and support more demanding tasks requiring precise object position or trajectory control. In addition, we will explore the applicability of the airflow manipulation mechanism to objects of different shapes and masses and to more complex scenarios.

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